

Homework Solutions

Mathematics

Student | 2026-10-08

Problem Q1

Suppose f is a function and L and a are real numbers.

Solution. See sub-problems for individual limit computations.

Answer:

See sub-problems

(a) $\lim_{x \rightarrow a} f(x) = L$ means: for every $\varepsilon > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$. Informally: we can make $f(x)$ as close to L as we like by taking x sufficiently close to (but not equal to) a . $\forall \varepsilon > 0, \exists \delta > 0 : 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$

(b) This requires graphing, which is a visual exercise. See graph

(c) This requires graphing, which is a visual exercise. See graph

(d) This requires graphing, which is a visual exercise. See graph

Problem Q2

Is ∞ a number? What does $\lim_{x \rightarrow a} f(x) = \infty$ mean? Does $\lim_{x \rightarrow a} f(x) = \infty$ mean that the limit exists?

Solution. ∞ is **not** a real number.

$\lim_{x \rightarrow a} f(x) = \infty$ means that $f(x)$ grows without bound as $x \rightarrow a$.

The limit does **not exist** in the usual sense; we say the limit “equals infinity” as a shorthand for divergence.

Answer:

∞ is not a number. The limit does not exist.

Problem Q3

What does the Squeeze Theorem say? Draw a picture illustrating this theorem.

Solution. Squeeze Theorem: If $g(x) \leq f(x) \leq h(x)$ for all x near a (except possibly at a), and $\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

Intuitively: if f is “squeezed” between two functions that both converge to L , then f must also converge to L .

Answer:

If $g(x) \leq f(x) \leq h(x)$ and $\lim g = \lim h = L$, then $\lim f = L$.

Problem Q4

What do we mean by $\lim_{x \rightarrow a^-} f(x) = L$? What is meant by $\lim_{x \rightarrow a^+} f(x) = L$? Graph a function $y = f(x)$ and some point $x = a$ where $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$. Find an algebraic expression for such a function.

Solution. $\lim_{x \rightarrow a^-} f(x) = L$: the **left-hand limit**. As x approaches a from values *less than* a , $f(x)$ approaches L .

$\lim_{x \rightarrow a^+} f(x) = L$: the **right-hand limit**. As x approaches a from values *greater than* a , $f(x)$ approaches L .

The two-sided limit $\lim_{x \rightarrow a} f(x) = L$ exists if and only if both one-sided limits exist and are equal.

Example: $f(x) = \begin{cases} x & x < 0 \\ x + 1 & x \geq 0 \end{cases}$ at $a = 0$: $\lim_{x \rightarrow 0^-} f(x) = 0 \neq 1 = \lim_{x \rightarrow 0^+} f(x)$.

Answer:

$\lim_{x \rightarrow a^-} f(x) : \text{from below;}$	$\lim_{x \rightarrow a^+} f(x) : \text{from above}$
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Problem P1

(a) Is the limit of a sum always the sum of the limits? Give an example where $\lim_{x \rightarrow a} [f(x) + g(x)]$ exists even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exist.

Solution. No, the limit of a sum is *not* always the sum of the limits.

The **Sum Law** requires both individual limits to exist: $\lim[f(x) + g(x)] = \lim f(x) + \lim g(x)$ if both limits exist.

Counterexample: Let $f(x) = \frac{1}{x}$ and $g(x) = -\frac{1}{x}$. Neither $\lim_{x \rightarrow 0} f(x)$ nor $\lim_{x \rightarrow 0} g(x)$ exists, but $\lim_{x \rightarrow 0} [f(x) + g(x)] = \lim_{x \rightarrow 0} 0 = 0$.

Answer:

No. $f(x) = 1/x$, $g(x) = -1/x$: $\lim[f + g] = 0$ but $\lim f, \lim g$ DNE

Problem P2

(a) Find $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$, $\lim_{x \rightarrow 0^-} (\frac{1}{x} - \frac{1}{|x|})$, and $\lim_{x \rightarrow 0} (\frac{1}{x} - \frac{1}{|x|})$, if they exist.

Solution. $\lim_{x \rightarrow 0^+} -\frac{1}{|x|} + \frac{1}{x} = 0$

$$\lim_{x \rightarrow 0^-} -\frac{1}{|x|} + \frac{1}{x} = -\infty$$

$$\lim_{x \rightarrow 0} -\frac{1}{|x|} + \frac{1}{x} = 0$$

Answer:

0

(b) Computing: $\lim_{x \rightarrow 0} |x| = 0$ This confirms the statement. $\square 0$

(c) Left limit: $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$ Right limit: $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$ Since $-1 \neq 1$, the limit does not exist. $\square \text{DNE}$

Problem P3

For what kinds of functions can one evaluate the limit $\lim_{x \rightarrow a} f(x)$ by just plugging in $x = a$? Give an example of a function $f(x)$ for which the limit $\lim_{x \rightarrow a} f(x)$ exists and can be evaluated, yet for which this substitution doesn't work.

Solution. **Continuous functions** allow direct substitution: $\lim_{x \rightarrow a} f(x) = f(a)$.

This includes polynomials, rational functions (at non-zero denominator points), trigonometric, exponential, and logarithmic functions at points in their domain.

Example where limit exists but substitution fails: $f(x) = \frac{x^2-1}{x-1}$ at $x = 1$. Direct substitution gives $\frac{0}{0}$, but $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = \lim_{x \rightarrow 1} (x+1) = 2$.

Answer:

Continuous functions: $\lim_{x \rightarrow a} f(x) = f(a)$

Problem AP1

Below are listed three limits. Match the best method to each limit and evaluate each limit.

Solution. See sub-problems for individual limit computations.

Answer:

See sub-problems

(a) $\lim_{x \rightarrow 0} \frac{x^{1000} - 703x^{506} + \pi x^{17} + 480}{-372x^{75} + 39x^{14} + \sqrt{7x^5 - 24}} = -20$ $\square -20$

(b) $\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4} = 10$ $\square 10$

(c) $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$ $\square 0$

Problem AP2

Using a calculator, try to guess $\lim_{n \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right)$. (Remember that n is in radians, not degrees!)

Solution. $\lim_{x \rightarrow \infty} n \sin\left(\frac{\pi}{n}\right) = \pi$

Answer:

π

Problem AP3

Draw a circle of radius 1. Inscribe a regular n -gon inside it.

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(a)

(b)

(c)

Auto-generated: 9/10 problems solved by SymPy.